TUTORIAL 2

Mathematical Language, Sets, and Proofs

2.1 The induction principle

DEFINITION 2.1

The principle of mathematical (weak) induction:
Let n_0 be a nonnegative integer, and let $P(n)$ be a statement defined for every integer $n \geq n_0$. Suppose that:

1. **Base case:** $P(n_0)$ is true.
2. **Inductive step:** for every integer $n \geq n_0$, the assumption $P(n)$ implies $P(n+1)$.

Then $P(n)$ is true for every integer $n \geq n_0$.

EXERCISE 2.1

Please prove by induction that $2 + 4 + \cdots + 2n = n(n+1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $2 = 1 \cdot 2$. Suppose $\sum_{j=1}^{k} 2j = k(k+1)$ for $k \geq 1$. Then

$$\sum_{j=1}^{k+1} 2j = k(k+1) + 2(k+1) = (k+1)(k+2).$$

Induction proves the identity for every $n \geq 1$.

We've got it. Victory!

EXERCISE 2.2

Please prove by induction that $17 \mid (18^n - 1)$ for every $n \geq 1$.

WORKED SOLUTION

For $n = 1$, $18 - 1 = 17$. Suppose $18^k - 1 = 17m$ for some $m \in \mathbb{Z}$. Then

$$18^{k+1} - 1 = 18(18^k - 1) + 17 = 17(18m + 1).$$

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EXERCISE 2.2 CONTINUED

This is divisible by 17, completing the induction step and the proof.

That does it. Victory!

EXERCISE 2.3

Please prove by induction that $4^n \geq 3n^2$ for every $n \geq 3$.

WORKED SOLUTION

For $n = 3$, $4^3 = 64 \geq 27 = 3 \cdot 3^2$. Suppose $4^k \geq 3k^2$ for $k \geq 3$. Then

$$4^{k+1} \geq 12k^2.$$

Since $4k^2 - (k+1)^2 = 3k^2 - 2k - 1 = (3k+1)(k-1) > 0$,

$$4^{k+1} \geq 12k^2 \geq 3(k+1)^2.$$

The induction proves the claim for every $n \geq 3$.

All wrapped up. Victory!

2.2 Sets

DEFINITION 2.2

- **Inclusion:** $A \subseteq B$ means that every element of A also belongs to B.
- **Union:** $A \cup B = \{\, x \mid x \in A \text{ or } x \in B \,\}$.
- **Intersection:** $A \cap B = \{\, x \mid x \in A \text{ and } x \in B \,\}$.

$\Omega = \{2,4,6,8,10\}, \quad A = \{2,4,6\}, \quad B = \{4,6,8\}$

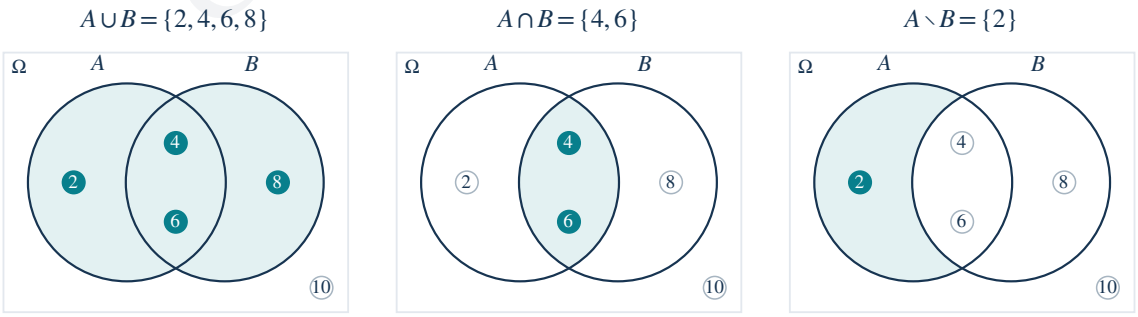


Figure 2.1. For $A = \{2,4,6\}$ and $B = \{4,6,8\}$ in $\Omega = \{2,4,6,8,10\}$, the shaded regions show $A \cup B$, $A \cap B$, and $A \setminus B$, respectively. Each numeral's position records its membership; 10 belongs to the universal set but to neither A nor B.

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EXERCISE 2.4

Please prove that $A \subseteq B$ if and only if $A \cup B = B$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. The definition of union gives $A \cup B \subseteq B$, while $B \subseteq A \cup B$ always holds. Therefore $A \cup B = B$.
Second direction: suppose $A \cup B = B$. Let $x \in A$. Then $x \in A \cup B$, so $x \in B$. Therefore $A \subseteq B$.

We've reached the finish. Victory!

EXERCISE 2.5

Please prove that $A \subseteq B$ if and only if $A \cap B = A$.

WORKED SOLUTION

First direction: suppose $A \subseteq B$. If $x \in A \cap B$, then $x \in A$ and $x \in B$, so $A \cap B \subseteq A$. Conversely, if $x \in A$, then the inclusion gives $x \in B$. Thus $x \in A \cap B$, proving $A \subseteq A \cap B$. Hence $A \cap B = A$.

Second direction: suppose $A \cap B = A$. Let $x \in A$. Then $x \in A \cap B$, so $x \in B$. Therefore $A \subseteq B$.

Yes! Victory!

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